

HW12

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Load setup and themes

I began by loading the required packages and setting consistent figure styles.

```
suppressPackageStartupMessages({  
  library(effects)  
  library(ggplot2)  
})  
knitr::opts_chunk$set(fig.width = 6.5, fig.height = 4.2, dpi = 300, fig.align = "center")  
theme_set(  
  theme_minimal(base_size = 11) +  
    theme(  
      panel.grid.minor = element_blank(),  
      plot.title = element_text(face = "bold"),  
      legend.position = "right"  
    )  
)
```

Load dataset

Here I import the dataset from the provided Dropbox link.

```
world <- read.csv("https://www.dropbox.com/scl/fi/1dzteukcsey51pjbfgf0b/world.csv?rlkey=pm9u4j")
```

Clean pr_sys variable

I ensure that the electoral system variable is labeled properly as “Non-PR” and “PR.”

```
if (!("pr_sys" %in% names(world))) stop("Column 'pr_sys' not found in dataset.")
if (!is.factor(world$pr_sys)) {
  vals <- sort(unique(na.omit(world$pr_sys)))
  if (all(vals %in% c(0, 1))) {
    world$pr_sys <- factor(world$pr_sys, levels = c(0, 1), labels = c("Non-PR", "PR"))
  } else {
    world$pr_sys <- factor(world$pr_sys)
  }
}
print(paste0("pr_sys levels: ", paste(levels(world$pr_sys), collapse = ", ")))
```

```
[1] "pr_sys levels: No, Yes"
```

Q1) Constant-only model

I start with a constant-only model to find the mean female representation.

```
model1 <- lm(women09 ~ 1, data = world)
summary(model1)
```

Call:

```
lm(formula = women09 ~ 1, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.177	-7.477	-1.627	5.773	39.123

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.1772	0.8238	20.85	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.05 on 179 degrees of freedom
(11 observations deleted due to missingness)

Q2) Model with per capita GDP

I add GDP per capita to test if wealthier countries have higher representation.

```
model2 <- lm(women09 ~ gdp_10_thou, data = world)
summary(model2)
```

Call:

```
lm(formula = women09 ~ gdp_10_thou, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-24.74	-6.74	-1.62	5.78	41.38

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.8430	0.9542	15.56	< 2e-16 ***
gdp_10_thou	3.4574	0.8351	4.14	5.5e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.38 on 167 degrees of freedom

(22 observations deleted due to missingness)

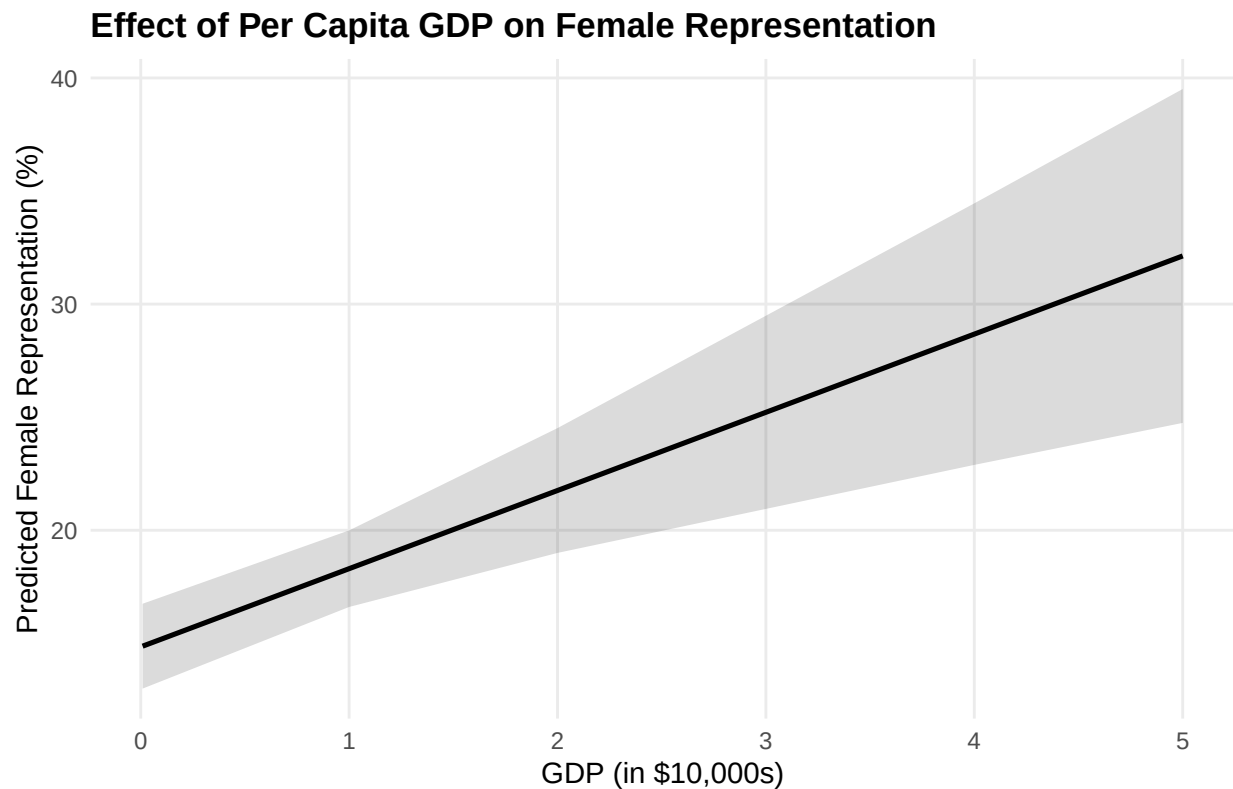
Multiple R-squared: 0.09308, Adjusted R-squared: 0.08765

F-statistic: 17.14 on 1 and 167 DF, p-value: 5.501e-05

Q3) Graph: Effect of per capita GDP

I visualize how GDP affects female representation with confidence bands.

```
df_eff2 <- as.data.frame(effects::Effect("gdp_10_thou", model2))
ggplot(df_eff2, aes(x = gdp_10_thou, y = fit)) +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.18) +
  geom_line(linewidth = 0.9) +
  labs(
    title = "Effect of Per Capita GDP on Female Representation",
    x = "GDP (in $10,000s)",
    y = "Predicted Female Representation (%)"
  )
```



Q4) Model with electoral system dummy

I now include electoral system type to test its independent effect.

```
model3 <- lm(women09 ~ pr_sys, data = world)
summary(model3)
```

Call:

```
lm(formula = women09 ~ pr_sys, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.589	-7.835	-1.860	7.140	33.911

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.1596	0.9686	14.619	< 2e-16 ***
pr_sysYes	8.2297	1.5996	5.145	7.02e-07 ***

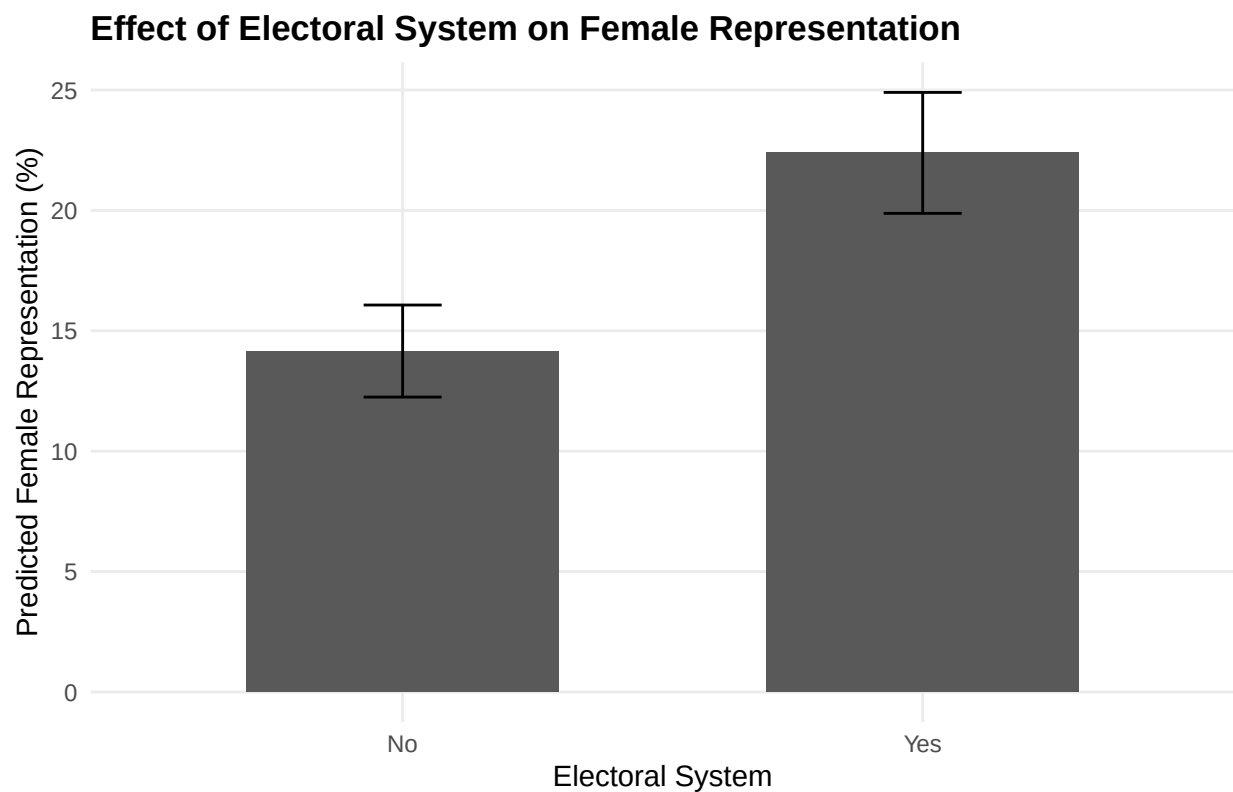
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.34 on 178 degrees of freedom
(11 observations deleted due to missingness)
Multiple R-squared: 0.1295, Adjusted R-squared: 0.1246
F-statistic: 26.47 on 1 and 178 DF, p-value: 7.019e-07

Q5) Graph: Effect of electoral system

I plot predicted values by system type.

```
df_eff3 <- as.data.frame(effects::Effect("pr_sys", model3))
ggplot(df_eff3, aes(x = pr_sys, y = fit)) +
  geom_col(width = 0.6) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  labs(
    title = "Effect of Electoral System on Female Representation",
    x = "Electoral System",
    y = "Predicted Female Representation (%)"
  )
)
```



Q6) Model with BOTH GDP and electoral system

I combine both predictors to see how they behave together.

```
model4 <- lm(women09 ~ gdp_10_thou + pr_sys, data = world)
summary(model4)
```

Call:

```
lm(formula = women09 ~ gdp_10_thou + pr_sys, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-20.371	-7.872	-1.266	6.399	36.148

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.3922	1.0248	12.092	< 2e-16 ***
gdp_10_thou	2.7864	0.7948	3.506	0.000585 ***
pr_sysYes	7.7007	1.5713	4.901	2.25e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.73 on 166 degrees of freedom

(22 observations deleted due to missingness)

Multiple R-squared: 0.2077, Adjusted R-squared: 0.1982

F-statistic: 21.76 on 2 and 166 DF, p-value: 4.049e-09

Q7) Compare model fit (R² and AIC)

I compare model performance to see which fits best.

```
print(c(
  R2_m1 = summary(model1)$r.squared,
  R2_m2 = summary(model2)$r.squared,
  R2_m3 = summary(model3)$r.squared,
  R2_m4 = summary(model4)$r.squared
))
```

R2_m1	R2_m2	R2_m3	R2_m4
0.0000000	0.0930778	0.1294601	0.2077099

```
AIC(model1, model2, model3, model4)
```

	df	AIC
model1	2	1378.788
model2	3	1274.428
model3	3	1355.832
model4	4	1253.592

Q8) Graph: GDP effect for PR countries only

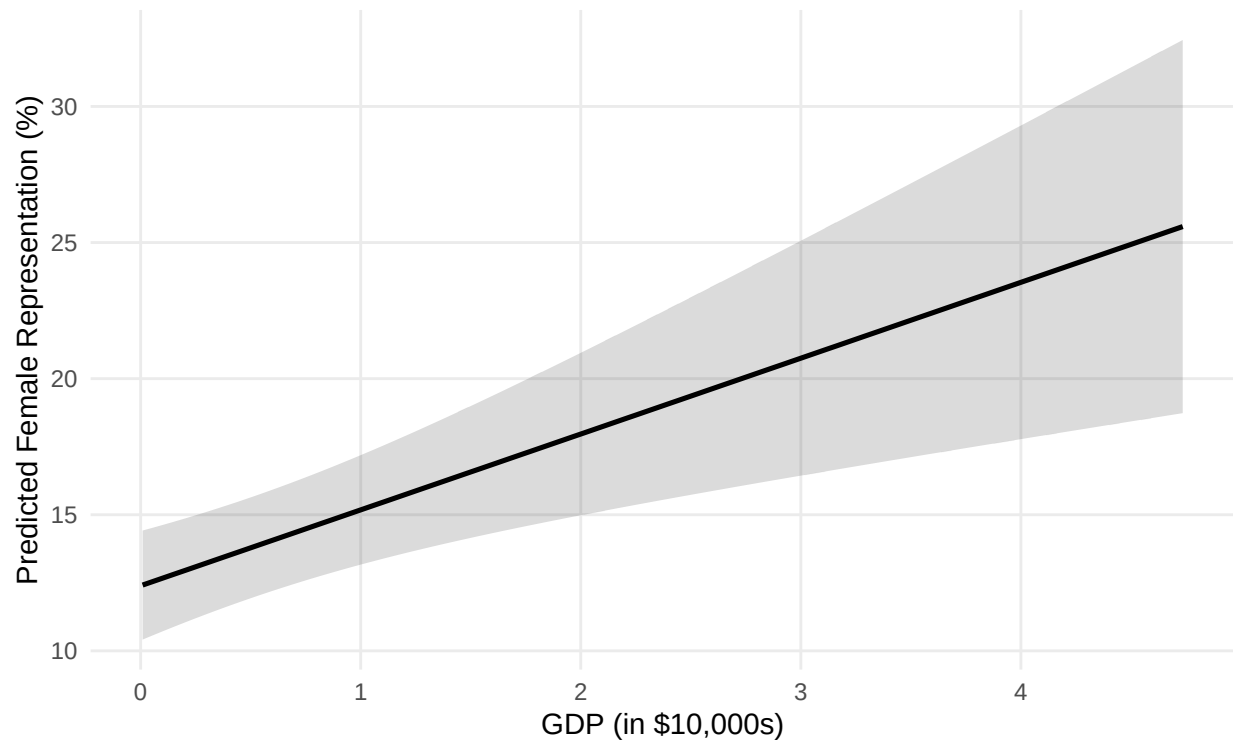
I show the relationship between GDP and representation within PR systems.

```
target_level <- if ("PR" %in% levels(world$pr_sys)) "PR" else levels(world$pr_sys)[1]
if (!("PR" %in% levels(world$pr_sys))) {
  print(paste0("Note: 'PR' not found. Using level '", target_level, "' instead. "))
}
```

```
[1] "Note: 'PR' not found. Using level 'No' instead."
```

```
gseq <- seq(min(world$gdp_10_thou, na.rm = TRUE),
            max(world$gdp_10_thou, na.rm = TRUE),
            length.out = 100)
nd_pr <- data.frame(
  gdp_10_thou = gseq,
  pr_sys = factor(target_level, levels = levels(world$pr_sys))
)
pred_pr <- predict(model4, newdata = nd_pr, se.fit = TRUE)
nd_pr$fit <- pred_pr$fit
nd_pr$lwr <- pred_pr$fit - 1.96 * pred_pr$se.fit
nd_pr$upr <- pred_pr$fit + 1.96 * pred_pr$se.fit
ggplot(nd_pr, aes(x = gdp_10_thou, y = fit)) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), alpha = 0.18) +
  geom_line(linewidth = 0.9) +
  labs(
    title = paste0("Effect of GDP on Female Representation (", target_level, " Only)"),
    x = "GDP (in $10,000s)",
    y = "Predicted Female Representation (%)"
  )
```

Effect of GDP on Female Representation (No Only)



Q9) Interaction (GDP × PR)

I test for an interaction to see if GDP's impact differs between PR and Non-PR systems.

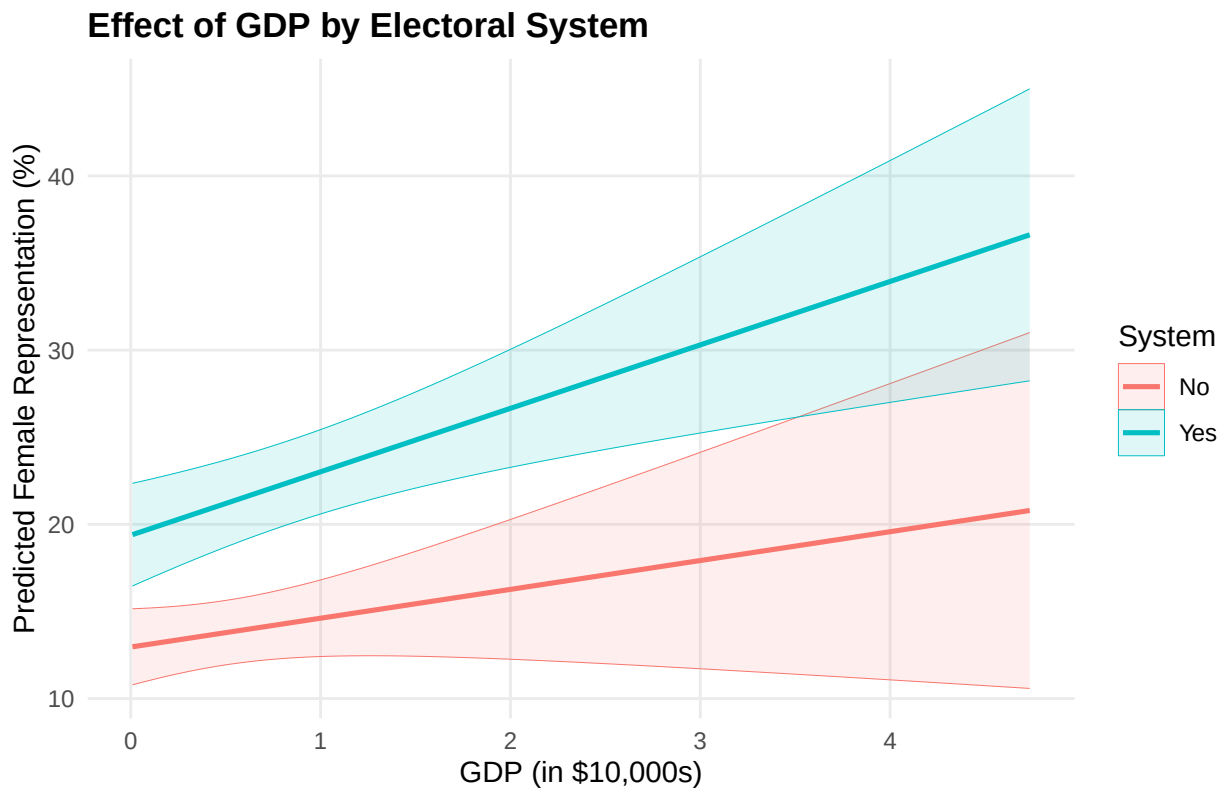
```
model_interact <- lm(women09 ~ gdp_10_thou * pr_sys, data = world)
gseq <- seq(min(world$gdp_10_thou, na.rm = TRUE),
            max(world$gdp_10_thou, na.rm = TRUE),
            length.out = 100)
nd_all <- expand.grid(
  gdp_10_thou = gseq,
  pr_sys = levels(world$pr_sys)
)
pred_all <- predict(model_interact, newdata = nd_all, se.fit = TRUE)
nd_all$fit <- pred_all$fit
nd_all$lwr <- nd_all$fit - 1.96 * pred_all$se.fit
nd_all$upr <- nd_all$fit + 1.96 * pred_all$se.fit
ggplot(nd_all, aes(x = gdp_10_thou, y = fit, color = pr_sys, fill = pr_sys)) +
  geom_ribbon(aes(ymin = lwr, ymax = upr), alpha = 0.12, linewidth = 0) +
  geom_line(linewidth = 0.9) +
  labs(
```



```

title = "Effect of GDP by Electoral System",
x = "GDP (in $10,000s)",
y = "Predicted Female Representation (%)",
color = "System", fill = "System"
)

```



Q10) Region model

I now test for regional variation in women's representation.

```

model_region <- lm(women09 ~ region, data = world)
summary(model_region)

```

Call:

```
lm(formula = women09 ~ region, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.984	-7.135	-1.746	4.914	38.740

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.5600	1.4445	12.157	< 2e-16 ***
regionAsia-Pacific	-5.0569	2.2407	-2.257	0.0253 *
regionC&E Europe	-0.7680	2.4171	-0.318	0.7511
regionMiddle East	-8.0705	2.6511	-3.044	0.0027 **
regionN. America	4.8067	5.7780	0.832	0.4066
regionS. America	0.4244	2.2407	0.189	0.8500
regionScandinavia	23.9600	4.5679	5.245	4.54e-07 ***
regionW. Europe	6.1926	2.6511	2.336	0.0207 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.69 on 172 degrees of freedom

(11 observations deleted due to missingness)

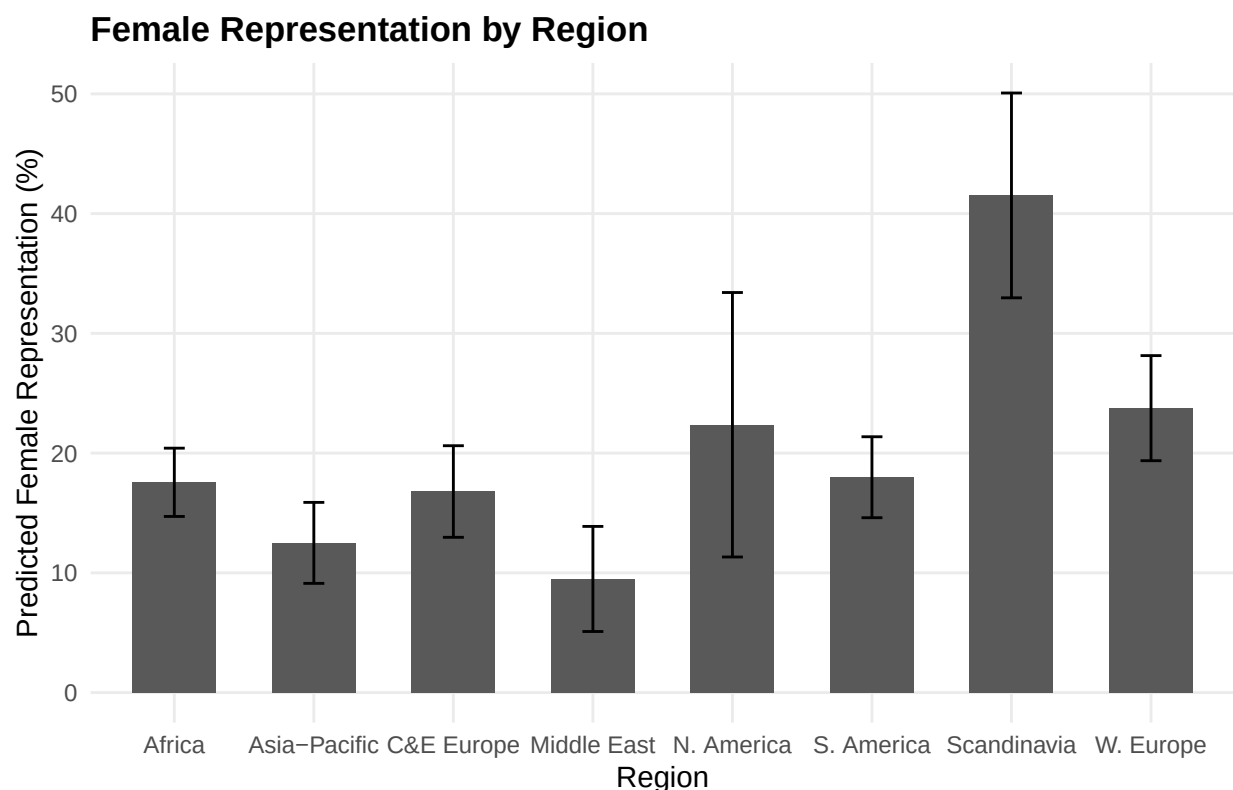
Multiple R-squared: 0.2615, Adjusted R-squared: 0.2314

F-statistic: 8.7 on 7 and 172 DF, p-value: 3.929e-09

Q11) Graph: Female representation by region

I plot predicted averages by region.

```
df_eff_region <- as.data.frame(effects::Effect("region", model_region))
ggplot(df_eff_region, aes(x = region, y = fit)) +
  geom_col(width = 0.6) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  labs(
    title = "Female Representation by Region",
    x = "Region",
    y = "Predicted Female Representation (%)"
  )
```



Q12) Model: GDP controlling for region

I control for GDP to isolate its impact within regional context.

```
model_gdp_region <- lm(women09 ~ gdp_10_thou + region, data = world)
summary(model_gdp_region)
```

Call:

```
lm(formula = women09 ~ gdp_10_thou + region, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.123	-6.959	-1.845	4.348	38.563

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.7141	1.4194	12.480	< 2e-16 ***
gdp_10_thou	1.0914	1.2707	0.859	0.391693
regionAsia-Pacific	-4.9372	2.2856	-2.160	0.032254 *
regionC&E Europe	-1.2315	2.3610	-0.522	0.602678

```

regionMiddle East  -11.0675      2.8079   -3.942  0.000121 ***
regionN. America    2.2842       6.1856    0.369  0.712410
regionS. America   -0.9546       2.2279   -0.428  0.668890
regionScandinavia  20.3981       5.8626    3.479  0.000648 ***
regionW. Europe     3.1906       3.9828    0.801  0.424264
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.38 on 160 degrees of freedom
(22 observations deleted due to missingness)
Multiple R-squared:  0.2903,    Adjusted R-squared:  0.2549
F-statistic: 8.183 on 8 and 160 DF,  p-value: 2.991e-09

```

Clean frac_eth3 variable

Before modeling ethnic fractionalization, I clean missing and blank entries.

```

if (!"frac_eth3" %in% names(world)) stop("Column 'frac_eth3' not found in dataset.")
KEEP_MISSING <- FALSE
MISSING_LABEL <- "Missing"
x <- as.character(world$frac_eth3)
is_blank <- !is.na(x) & nchar(trimws(x)) == 0
x[is_blank] <- NA_character_
if (KEEP_MISSING) {
  x[is.na(x)] <- MISSING_LABEL
  world$frac_eth3 <- factor(x)
  print("Kept missing/blank values as labeled.")
} else {
  keep <- !is.na(x)
  dropped <- sum(!keep)
  world <- world[keep, , drop = FALSE]
  world$frac_eth3 <- factor(x[keep])
  print(paste("Dropped", dropped, "rows with missing/blank frac_eth3."))
}

```

```
[1] "Dropped 3 rows with missing/blank frac_eth3."
```

```
print(paste("Final frac_eth3 levels:", paste(levels(world$frac_eth3), collapse = ", ")))
```

```
[1] "Final frac_eth3 levels: High, Low, Medium"
```

Q13) Ethnic fractionalization model

I estimate representation across ethnic fractionalization categories.

```
model_ethnic <- lm(women09 ~ frac_eth3, data = world)
summary(model_ethnic)
```

Call:

```
lm(formula = women09 ~ frac_eth3, data = world)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.457	-7.457	-1.536	5.343	39.416

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.43559	1.44206	12.091	<2e-16 ***
frac_eth3Low	0.02178	2.02259	0.011	0.991
frac_eth3Medium	-0.55138	2.05719	-0.268	0.789

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.08 on 174 degrees of freedom

(11 observations deleted due to missingness)

Multiple R-squared: 0.000573, Adjusted R-squared: -0.01091

F-statistic: 0.04988 on 2 and 174 DF, p-value: 0.9514

Q14) Graph: Effect of ethnic fractionalization

I visualize predicted representation by fractionalization level.

```
eff_eth <- effects::Effect("frac_eth3", model_ethnic)
df_eff_eth <- as.data.frame(eff_eth)
ggplot(df_eff_eth, aes(x = frac_eth3, y = fit)) +
  geom_col(width = 0.6) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  geom_text(aes(label = sprintf("%.1f", fit)), vjust = -0.5, size = 3.3) +
  labs(
    title = "Effect of Ethnic Fractionalization on Female Representation",
    x = "Ethnic Fractionalization Level",
    y = "Predicted Female Representation (%)"
  )
```

```
) +  
ylim(0, max(df_eff_eth$upper, na.rm = TRUE) * 1.1)
```

